Figure 12.10 The blockchain butterfly: foundations to contributions

12.13 Looking ahead

At the global level, organizations such as the United Nations have an opportunity to serve as a test bed for many blockchain applications, such as verifiable disaster relief, vaccine distribution, maintaining international peace through trust, and enforcing democratic processes. Chapters 7 and 8 demonstrate a massive global plastics cleanup problem.

Within any country, government officials and policymakers can take up policies and regulations to ease the adoption of blockchain. U.S. states such as New York and Delaware are considering regulations that will help in the widespread adoption of blockchain solutions. Unlike with other traditional technologies, the government can host a full node or nodes to facilitate decentralized operations. This feature offers the additional benefit of full nodes storing a completed timestamped ledger of Txs. The information on the ledger can be used for audit and review purposes. The Department of Education, for example, may play a vital leadership role in the adoption of blockchain for educational purposes such as countrywide credentialing. (Chapter 11 discusses the credentialing prototype DCC-Dapp.)

At the application level, autonomous vehicles and robots on shop floors and in home care have become a reality. Augmenting these innovations with blockchain can develop a layer of trust for monitoring autonomous actions. Enabling a cryptocurrency capability for these autonomous entities, you can design self-paying and self-managing machines. You can send